Customer Segmentation using K-Means Clustering

Aim:
To analyze customer purchasing behavior and divide customers into different groups using the K-Means clustering algorithm based on annual income and spending score.

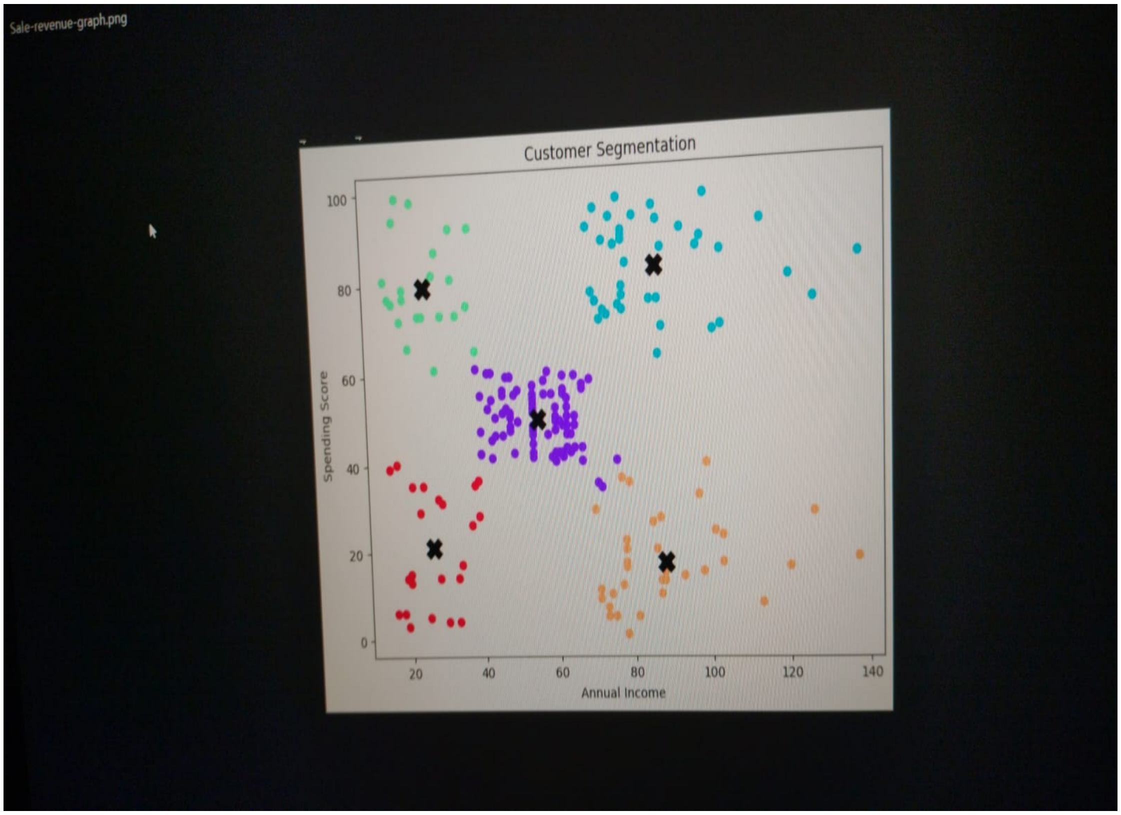
Objectives:
• To understand customer behavior using data analysis.
• To apply machine learning techniques for customer segmentation.
• To identify groups of customers with similar spending habits.
• To help businesses improve marketing strategies and customer targeting.
• To visualize customer groups using graphical representation.

Introduction:
Customer segmentation is a process of dividing customers into different groups based on common characteristics such as income, purchasing behavior, and spending patterns. Businesses use customer segmentation to understand customer needs and improve marketing strategies. In this project, the K-Means clustering algorithm is used to group customers according to their Annual Income and Spending Score.

Dataset Information	Details
Dataset Name	Mall Customers Dataset
Total Records	200
Total Columns	5
Columns Used	Annual Income (k$), Spending Score (1-100)

Tools and Technologies Used:
• Python Programming Language
• Google Colab / Jupyter Notebook
• Pandas Library for data analysis
• Matplotlib for visualization
• Scikit-learn for K-Means clustering

Methodology:
1. The dataset was loaded into Python using Pandas.
2. Important columns such as Annual Income and Spending Score were selected.
3. The K-Means clustering algorithm was applied to divide customers into clusters.
4. Five customer groups were created using K-Means.
5. A scatter plot was generated to visualize customer segmentation.
6. Cluster centers were identified using centroid points.



Cluster Interpretation:
• Cluster 1: Customers with low income and low spending score.
• Cluster 2: Customers with high income and high spending score.
• Cluster 3: Average customers with medium spending habits.
• Cluster 4: Customers with high income but low spending score.
• Cluster 5: Customers with low income but high spending score.

Advantages of Customer Segmentation:
• Helps businesses understand customer behavior.
• Improves marketing and advertisement strategies.
• Increases customer satisfaction.
• Helps in personalized product recommendations.
• Improves business decision making.

Result:
The K-Means clustering algorithm successfully grouped customers into five different segments based on their annual income and spending score. The graph clearly shows distinct customer groups and their purchasing behavior.

Conclusion:
Customer segmentation using K-Means clustering is an effective method for analyzing customer behavior. This project demonstrates how machine learning techniques can help businesses identify valuable customer groups and improve marketing strategies. The project also provides useful visual insights into customer spending patterns.

Future Scope:
• Use more customer attributes for better segmentation.
• Apply advanced machine learning algorithms.
• Create a real-time dashboard using Power BI or Tableau.
• Integrate customer segmentation into e-commerce platforms.